

$$\begin{aligned}
& d) [(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r \\
& \equiv \neg [(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \vee r \quad (\text{By conditional-disjunction equivalence}) \\
& \equiv [\neg(p \vee q) \vee \neg(p \rightarrow r) \vee \neg(q \rightarrow r)] \vee r \quad (\text{By 1st De Morgan's Law}) \\
& \equiv [\neg(p \vee q) \vee \neg(p \rightarrow r) \vee \neg(q \rightarrow r)] \vee r \quad (\text{By 2nd De Morgan's Law}) \\
& \equiv [\neg(p \vee q) \vee \neg(\neg p \vee r) \vee \neg(\neg q \vee r)] \vee r \quad (\text{By 2 applications of conditional-disjunction equivalence}) \\
& \equiv [\neg(p \vee q) \vee (\neg(\neg p) \wedge \neg r) \vee (\neg(\neg q) \wedge \neg r)] \vee r \quad (\text{By 2 applications of 2nd De Morgan's Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r) \vee (q \wedge \neg r)] \vee r \quad (\text{By 2 applications of Double negation Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r)] \vee [q \wedge \neg r \vee r] \quad (\text{By 1st Associative Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r)] \vee [r \vee (q \wedge \neg r)] \quad (\text{1st Commutative Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r)] \vee [(p \vee q) \wedge (p \vee \neg r)] \quad (\text{1st Distributive Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r)] \vee [(p \vee q) \wedge T] \quad (\text{1st Negation Law}) \\
& \equiv [\neg(p \vee q) \vee (p \wedge \neg r)] \vee (p \vee q) \quad (\text{1st Identity Law}) \\
& \equiv (\neg(p \vee q) \vee [(p \wedge \neg r) \vee (p \vee q)]) \quad (\text{1st Associative Law}) \\
& \equiv (\neg(p \vee q) \vee [(p \wedge \neg r) \vee q]) \quad (\text{1st Associative Law}) \\
& \equiv (\neg(p \vee q) \vee [r \vee (p \wedge \neg r)]) \quad (\text{1st Commutative Law}) \\
& \equiv (\neg(p \vee q) \vee [(p \vee r) \wedge (p \vee \neg r)]) \quad (\text{1st Distributive Law}) \\
& \equiv (\neg(p \vee q) \vee [(p \vee r) \wedge T]) \quad (\text{1st Negation Law}) \\
& \equiv (\neg(p \vee q) \vee (p \vee r)) \quad (\text{1st Identity Law}) \\
& \equiv \neg(p \vee q) \vee [(p \vee r) \vee q] \quad (\text{2nd De Morgan's Law}) \\
& \equiv \neg(p \vee q) \vee [r \vee (p \vee q)] \quad (\text{1st Associative Law}) \\
& \equiv \neg(p \vee q) \vee [(p \vee q) \vee r] \quad (\text{1st Commutative Law}) \\
& \equiv [(\neg(p \vee q) \vee (p \vee q))] \vee r \quad (\text{1st Associative Law}) \\
& \equiv [(\neg(p \vee q) \vee \neg(p \vee q))] \vee r \quad (\text{1st Commutative Law}) \\
& \equiv T \vee r \quad (\text{1st Negation Law}) \\
& \equiv r \vee T \quad (\text{1st Commutative Law}) \equiv T \quad (\text{1st Domination Law})
\end{aligned}$$